

Theorem 1.4 (Weyl). *Let $p(n) = a_k n^k + \cdots + a_0$ be a real polynomial with at least one coefficient among $a_1, \dots, a_k$ irrational. Then the sequence $(p(n))$ is equidistributed modulo 1.*

Furstenberg extended unique ergodicity to a dynamically defined extension of the irrational circle rotation described in Example 1.3, giving an elegant ergodic-theoretic proof of Theorem 1.4. This approach will be discussed in Sect. 4.4.

Exercises for Sect. 1.2

Exercise 1.2.1. Describe what Theorem 1.4 can tell us about the leading digits of the powers of 2.

1.3 Szemerédi's Theorem

Szemerédi, in an intricate and difficult combinatorial argument, proved a long-standing conjecture of Erdős and Turán [85] in his paper [357]. A set S of integers is said to have *positive upper Banach density* if there are sequences (m_j) and (n_j) with $n_j - m_j \rightarrow \infty$ as $j \rightarrow \infty$ with the property that

$$\lim_{j \rightarrow \infty} \frac{|S \cap [m_j, n_j]|}{n_j - m_j} > 0.$$

Theorem 1.5 (Szemerédi). *Any subset of the integers with positive upper Banach density contains arbitrarily long arithmetic progressions.*

Furstenberg [102] (see also his book [103] and the article of Furstenberg, Katznelson and Ornstein [107]) showed that Szemerédi's theorem would follow from a generalization of Poincaré's recurrence theorem, and proved that generalization. The connection between recurrence and Szemerédi's theorem will be explained in Sect. 7.3, and Furstenberg's proof of the generalization of Poincaré recurrence needed will be presented in Chap. 7. There are a great many more theorems in this direction which we cannot cover, but it is worth noting that many of these further theorems to date only have proofs using ergodic theory.

More recently, Gowers [122] has given a different proof of Szemerédi's theorem, and in particular has found the following effective form of it*.

Theorem (Gowers). For every integer $s \geq 1$ and sufficiently large integer N , every subset of $\{1, 2, \dots, N\}$ with at least

* Theorems and other results that are not numbered will not be proved in this volume, but will also not be used in the main body of the text.